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PAPER_1006: ALICE Multiplicity SCm Phonon Scaling

Abstract

We model the ALICE Pb-Pb charged-particle multiplicity $dN_{ch}/d\eta$ at $\sqrt{s_{NN}} = 5.02$ TeV with SCm phonon scaling. The $S_{26}^{(3)}$ phonon coupling enhances the base multiplicity by $(1 + \Phi)$.

1. Multiplicity Formula

$$dN_{ch}/d\eta = \alpha \cdot (N_{part}/2)^\beta \cdot (1 + \Phi) \cdot (\sqrt{s_{NN}}/200)^{0.15}$$

With $\alpha = 8.5$, $\beta = 1.25$, $N_{part} = 383$ (central 0-5%): $dN_{ch}/d\eta \approx 7565$.

2. SCm Enhancement

The phonon correction $\Phi = S_{26}^{(3)} = 0.095$ provides a 9.5% enhancement over the baseline empirical fit.

3. Implementation

File: scm_qgp_dynamics.py, class ALICEMultiplicityCalc. CP4 class #590.